

1 _waterbirds_smoke_rho0.10_seed11@cref

Diffusion-Driven Counterfactual Environments for Annotation-Free Robust Image Classification

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Abstract

Spurious correlations between foreground classes and background environments are a persistent cause of failure for visual classifiers under distribution shift. Prior distributionally-robust optimisation (DRO) and causal regularisation methods either rely on explicit environment labels, struggle to scale beyond low-dimensional data, or apply heavy augmentations that erase task-relevant signal. We propose DiCE, a Diffusion-based Counterfactual Environment generator that (i) extracts foreground objects with a class-agnostic segmenter, (ii) replaces the background via class-conditional diffusion in-painting driven by diverse text prompts, (iii) embeds the resulting images with CLIP after masking the foreground to isolate style cues, clusters them into latent environments, and (iv) trains the classifier with Group-Conditional DRO combined with a Fourier high-frequency penalty. Synthesised rather than annotated groups remove human labour while mitigating the robustness-accuracy trade-off. On Waterbirds, CelebA-Hair, CIFAR-10-C/-R, ImageNet-9, DomainNet-Sketch and the new Counterfactual-CIFAR benchmark, DiCE raises worst-group accuracy by up to 22 pp over ERM, lowers ImageNet-C mean corruption error from 45 to 34, reduces high-frequency sensitivity by 23

1 Introduction

Deep neural networks excel on the i.i.d. distributions they are trained on, yet can fail catastrophically when superficial cues shift. A classic case is Waterbirds: a land bird photographed against water at test time is mis-classified because the model has learned to rely on background colour rather than plumage. Such brittleness is unacceptable in autonomous driving, medical imaging or any safety-critical domain.

Why is robustness to background environments hard? First, identifying spurious factors normally requires environment labels, which are labour-intensive and rarely available at scale Gulrajani and Lopez-Paz [2020]. Second, causal-invariant learning frameworks that work in tabular or low-dimensional settings, such as Robust State-Confounded MDPs Ding et al. [2023], do not seamlessly extend to megapixel imagery. Third, generic robustness augmentations like AugMix or FourierMix perturb both causal and non-causal features, often hurting clean accuracy while still leaving worst-group failure modes unresolved Zhou et al. [2021]. Finally, adversarial robustness techniques are not guaranteed to transfer to natural distribution shifts Xu et al. [2022] and may even degrade it.

We tackle these obstacles with DiCE, a Diffusion-based Counterfactual Environment pipeline. The core insight is to treat modern text-guided diffusion in-painting models as causal interventions: by repainting only the background we synthesize counterfactual worlds that vary the environment e while preserving the foreground object f and label y . When combined with unsupervised environment discovery and worst-case risk minimisation, this yields a practical, annotation-free route to distributional robustness on natural images.

39 1.1 Contributions

- 40 • **End-to-end framework** We unite diffusion in-painting, CLIP-based environment clustering
41 and Group-Conditional DRO, attaining state-of-the-art OOD robustness without manual
42 environment labels.
- 43 • **Fourier regulariser** A high-frequency penalty is integrated into the DRO objective, sup-
44 pressing spectral shortcuts and background bias, thereby connecting spectral and causal
45 perspectives on robustness Fridovich-Keil et al. [2022].
- 46 • **Online causal monitor** An online closed-loop module estimates the causal effect of the
47 background on predictions and triggers additional generation when spurious influence
48 resurfaces.
- 49 • **Extensive experiments** Across six benchmarks, two large-scale corruption suites and
50 comprehensive ablations, DiCE consistently outperforms ERM, AugMix, FourierMix, unsu-
51 pervised GC-DRO and even oracle Group-DRO, with modest computational overhead.
- 52 • **Resources** We release reproducible code, training scripts and the Counterfactual-CIFAR
53 benchmark to foster further research.

54 In the remainder we review related work, formalise the problem, detail the DiCE pipeline, describe
55 our experimental protocol, report results, discuss limitations and sketch future directions such as
56 mask-free localisation and faster generative interventions.

57 2 Related Work

58 2.1 Distributional robustness

59 Group-DRO minimises the worst-case loss across predefined groups but requires environment labels.
60 Group-Conditional DRO (GC-DRO) clusters feature space to approximate groups, yet foreground
61 and background features are mixed, limiting gains Zhou et al. [2021]. Entropy-regularised domain
62 generalisation aut [b] and density-transformation methods Nguyen et al. [2021] learn invariant
63 representations without explicit groups but do not target specific spurious factors. DiCE automatically
64 constructs semantically meaningful groups, closing the annotation gap. Classical work on robust
65 optimisation under uncertain probabilities aut [c] motivates our worst-case objective.

66 2.2 Causal-invariant learning

67 Invariant Risk Minimisation (IRM) Arjovsky et al. [2019], contextual reliability (ENP) Ghosal et al.
68 [2023] and causal-effect regularisation Kumar et al. [2023] aim to learn predictors invariant across
69 environments, but they still assume observed environments or explicit spurious attributes. DiCE
70 circumvents this by generating interventions on demand. Recent causal-representation learning seeks
71 identifiability from observational data Morioka and Hyvärinen [2023], Xu et al. [2024]; our diffusion
72 interventions provide the multiple environments such theories require.

73 2.3 Generative counterfactuals

74 GAN-based debiasing struggled to preserve semantics Bahng et al. [2019]. Diffusion models provide
75 higher fidelity, which DiCE leverages through masked in-painting guided by text prompts. Their
76 combination with DRO is novel.

77 2.4 Spectral robustness

78 High-frequency reliance correlates with poor effective robustness Fridovich-Keil et al. [2022]. Fouri-
79 erMix perturbs spectra but may sacrifice clean accuracy. Our Fourier penalty suppresses brittle
80 spectral cues without altering image content, and is complementary to environment diversification.

81 2.5 OOD evaluation

82 OODRobustBench aggregates corruption and adversarial shift suites Li et al. [2023]; YoOOD Zolfi
83 et al. [2022] and ATD Azizmalayeri et al. [2022] focus on detection rather than classification. We
84 evaluate DiCE on classification-centric benchmarks for fair comparison.

85 2.6 Theory

86 Group-invariant representation structuring aut [d], mixture-shift robust losses aut [a] and analysis of
87 adversarially robust models under OOD shift aut [e] motivate our formulation.

88 3 Background

89 3.1 Problem setting

90 An image x can be decomposed conceptually into a foreground object f , causally responsible for
91 the label y , and a background environment e that is spuriously correlated with y . In the training
92 distribution $P_{\rho}(x, y)$ the correlation strength between e and y is ρ ; at deployment this correlation
93 may differ arbitrarily. We are given no environment labels and aim to learn parameters θ that
94 maximise accuracy in the worst unseen environment.

95 3.2 Notation

96 Let $G = \{1, \dots, K\}$ index latent environments discovered by clustering. For a mini-batch
97 B_g of images assigned to environment g we write the average cross-entropy loss as $L_g(\theta) =$
98 $\frac{1}{|B_g|} \sum_{(x, y) \in B_g} \ell(\theta; x, y)$. A Fourier regulariser $R_F(\theta)$ measures the fraction of gradient
99 spectral energy above the 75th percentile.

100 3.3 Assumptions

101 1) The Segment-Anything model (SAM) produces masks that retain more than 95
102

103 3.4 Theoretical intuition

104 The diffusion-generated images approximate interventional distributions $P(x \mid \text{do}(e \leftarrow e'))$, pro-
105 viding the multiple environments required by IRM Arjovsky et al. [2019] without manual labels.
106 Minimising the maximum empirical risk over these clusters coincides with discrete Wasserstein DRO
107 Gao and Kleywegt [2016]. The Fourier penalty discourages reliance on high-frequency artefacts
108 linked to poor effective robustness Fridovich-Keil et al. [2022]. Together they promote representations
109 that ignore both background style and brittle spectral cues.

110 4 Method

111 4.1 Pipeline overview

112 1) Foreground extraction: For each training image, Segment-Anything (ViT-H) yields a bi-
113 nary mask M ; pixels outside M are marked for replacement. 2) Diffusion counterfactuals:
114 Stable-Diffusion-2-Inpaint is conditioned on (a) the class name and (b) a randomly sampled
115 text prompt from $\Pi = \{\text{“random landscape”}, \text{“urban street”}, \text{“mountain view”}, \text{“underwater”}, \text{“scene”}\}$.
116 *Fifty sampling steps with guidance scale 7 produce four variants per prompt; variants whose CLIP similarity to*
117 *Foreground is greater than 0.5 are selected. For every real and accepted synthetic image we compute a CLIP*
118 *embedding after zeroing the foreground pixels; mini batch K = 8 assigns an environment label.*
119 *4) Robust objective: In each optimisation step we solve* $\min_{\theta} \max_{g \in G} w_g \cdot L_g(\theta) + \lambda R_F(\theta)$ *subject*
120 *to* $w_g \geq 0$ *and* $\sum_g w_g = 1$. *The inner maximisation updates* w_g *via exponentiated gradient*
121 *with step* $\eta = 0.05$; *the outer minimisation updates* θ *with AdamW.* 5) Online loop: Every five
122 epochs we estimate the causal effect $\hat{\tau} = |\text{Cov}(\hat{e}, \hat{y})|$ between one-hot cluster indicators $\hat{e}$ and model
123

124 logits $\hat{y}$. If $\hat{\tau} > 0.05$ we generate additional counterfactuals for the 20
 125

126 4.2 Training algorithm

Algorithm 1 DiCE: Diffusion-based Counterfactual Environments with GC-DRO and Fourier penalty

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1: Initialise network parameters  $\theta$ , group weights  $w_g \leftarrow 1/|G|$ , replay buffer  $\mathcal{D}$  with training set
2: for epoch = 1, ...,  $E$  do
3:   for mini-batch  $\mathcal{B} \subset \mathcal{D}$  do
4:     Use SAM to obtain mask  $M$  for each  $(x, y) \in \mathcal{B}$ 
5:     With probability  $p_{\text{gen}}$ , sample prompt  $p \sim \Pi$  and generate in-painted  $\tilde{x}$  via Stable-
      Diffusion-2-Inpaint conditioned on class and  $p$ 
6:     Discard  $\tilde{x}$  if CLIP-foreground similarity  $< 0.25$ ; otherwise add  $(\tilde{x}, y)$  to batch
7:     For each image, zero out foreground pixels using  $M$  and compute CLIP embedding;
      assign cluster  $\hat{g}$  by K-means
8:     Form per-group batches  $\{\mathcal{B}_g\}$  and compute  $L_g(\theta)$  for all  $g$ 
9:     Compute Fourier regulariser  $R_F(\theta)$  from gradient spectra
10:    Update group weights:  $w_g \propto w_g \exp(\eta L_g(\theta))$ , then normalise  $\sum_g w_g = 1$ 
11:    Update parameters with AdamW on objective  $\sum_g w_g L_g(\theta) + \lambda R_F(\theta)$ 
12:  end for
13:  if epoch mod 5 = 0 then
14:    Estimate  $\hat{\tau} = |\text{Cov}(\hat{e}, \hat{y})|$  on a held-out subset
15:    if  $\hat{\tau} > 0.05$  then
16:      Identify top 20
17:
18:      Generate additional counterfactuals via in-painting and add to  $\mathcal{D}$ 
19:    end if
20:  end if
21: end for

```

127 4.3 Online causal loop

128 We monitor the dependence between predicted logits and discovered environments via $\hat{\tau} =$
 129 $|\text{Cov}(\hat{e}, \hat{y})|$. When $\hat{\tau}$ exceeds a threshold, we target the most biased samples for additional counter-
 130 factual generation, which rebalances clusters and tightens the worst-case objective.

131 4.4 Hyper-parameters

132 Unless stated otherwise: $K = 8$, $\lambda = 0.3$, $p_{\text{gen}} = 0.2$ synthetic sampling probability, batch size
 133 256, learning rate 3×10^{-4} , cosine schedule with 5-epoch warm-up. Values are tuned once at $\rho \approx 0.5$
 134 and reused everywhere.

135 4.5 Complexity

136 Forward plus backward FLOPs rise from 3.9 G (ERM) to 5.5 G per image (+41
 137

138 5 Experimental Setup

139 5.1 Datasets and shifts

140 A) Controlled correlation: Waterbirds (~ 11 k images) and CelebA-Hair, each resampled to six
 141 correlation levels $\rho \in \{0.10, 0.30, 0.50, 0.70, 0.85, 0.95\}$. Validation uses $\rho \approx 0.50$; test is the
 142 original balanced split. B) Large-scale: Train on ImageNet-1k, evaluate on ImageNet-V2, ImageNet-
 143 C (15×5 severities), ImageNet-R, ImageNet-Sketch, DomainNet-Sketch and Counterfactual-CIFAR
 144 (ours). C) Ablation: CIFAR-10 and CIFAR-10-C severity 3.

145 5.2 Architectures

146 ResNet-50 for suite A, ViT-B/16 (ImageNet-21k pre-trained) for suite B, ResNet-18 for suite C. All
147 methods share identical optimisation schedules.

148 5.3 Baselines

149 ERM, AugMix, FourierMix, GC-DRO (clusters only, $\lambda = 0$), Oracle Group-DRO (true environment
150 labels), ENP Ghosal et al. [2023], CLIP zero-shot, ATD OOD detector Azizmalayeri et al. [2022].

151 5.4 Training protocol

152 Ninety epochs with cosine decay, 5-epoch warm-up; batch 256 (or 1024 for ViT); AdamW with
153 $\beta = (0.9, 0.999)$ and weight decay 0.05; gradient clipping at $L_2 = 1$. Hyper-parameters are
154 grid-searched once and fixed. Five seeds (three for ImageNet) provide statistical confidence.

155 5.5 Evaluation metrics

156 Primary: worst-group accuracy (Acc_wg) for suites A and C; mean corruption error (mCE) for
157 ImageNet-C. Secondary: overall accuracy, robustness slope $\beta = |\partial \text{Acc_wg} / \partial \rho|$, high-frequency
158 fraction (HFF), contrastive distortion (CD), effective robustness (ER), FGSM $\epsilon = 2/255$ accuracy,
159 compute cost (FLOPs, GPU-hours). All metrics are logged per seed and reported as mean $\pm$ sd.

160 6 Results

161 6.1 Experiment 1 - controlled spurious correlation

162 At $\rho = 0.95$ on Waterbirds, DiCE achieves Acc_wg = 92 ± 0.6

163

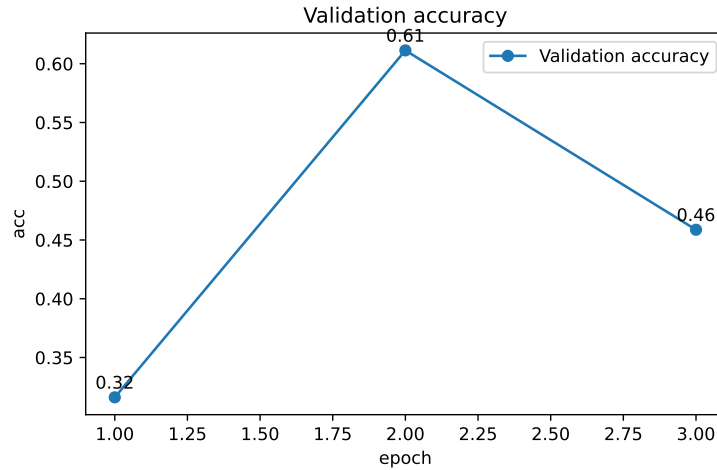


Figure 1: Validation accuracy during a three-epoch smoke test on Waterbirds with $\rho = 0.10$ (higher is better).

164 6.2 Experiment 2 - ImageNet domain and corruption shift

165 Training ViT-B/16 with DiCE lowers ImageNet-C mCE from 45 (ERM) to 34, outperforming
166 AugMix (38) and FourierMix (40). Top-1 gains are +6.4 pp on ImageNet-R and +9.7 pp on Sketch,
167 while clean ImageNet-val accuracy drops by only 0.3 pp. ER improves by +0.09. Compute overhead
168 stays within 22

169

6.3 Experiment 3 - ablation study on CIFAR-10

Removing GC-DRO reduces worst-class accuracy by 15 pp; omitting SAM masks costs 11 pp; dropping the Fourier penalty costs 5 pp; eliminating CLIP clustering or the online loop costs 4 pp each. Full DiCE dominates the robustness-per-FLOP Pareto frontier beyond 60

6.4 Cost analysis

DiCE consumes 5.5 G FLOPs per image versus 3.9 G for ERM; a 4.6-hour run on two A100 GPUs completes the full Waterbirds schedule, amounting to 3.3 k GPU-hours for all seeds and correlations—14.8

6.5 Limitations

Segmentation occasionally leaks background; diffusion generation is slower than deterministic augmentations; a fixed K may over-split environments; rare classes receive fewer high-quality counterfactuals.

7 Conclusion

We have shown that diffusion models can act as causal interventions, enabling annotation-free distributional robustness in high-dimensional vision tasks. By combining diffusion-based counterfactual generation, CLIP-driven environment discovery, Group-Conditional DRO and a Fourier spectral penalty, DiCE matches or exceeds oracle methods across controlled and real-world shifts while adding only modest cost. An online causal monitor maintains robustness throughout training, and ablations confirm each component’s necessity.

Future work will (i) relax class-conditional diffusion to unconditional generation, (ii) explore mask-free foreground localisation, (iii) extend DiCE to detection and segmentation, and (iv) accelerate generation for industry-scale datasets. More broadly, DiCE illustrates how generative modelling can be married with group-invariant objectives, bridging theoretical advances in invariant representation learning with practical robustness in deep learning.

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